Information to students

Dear student, welcome to Industrial Project Management 1, Exam

Time: 3 hours only

Exam instructions

1. The exam consists of 30 questions. All questions are compulsory.

2. Each question is worth one point, leading to a total score of 30 points.

3. Permitted aids: You are allowed to bring a simple calculator during this exam, and an English translating dictionary in book form is allowed. All other notes or books are NOT allowed. This is an individual exam and closed book exam.

4. The exam can be navigated at the bottom of the page, so if you are not sure about an answer, you may bookmark it and return to it later.

5. Remember to save it before you submit it.

6. To pass the exam, you will need at least 15 points

Formulas that might be useful

- PERT: Expected time = (O + 4M + P) / 6
- Variance= $(P-O/6)^2$
- SPI = EV / PV.
- CPI = EV/AC

Grading

5: 27–30
4: 23–26
3: 15–22
U: 0–14

wish you the best of luck

1 New Question

Which of the following dimensions is **NOT** part of the iron triangle ?
Select one alternative:

- Time
- Stakeholders ✔
- Quality
- Resources (cost)

Maximum marks: 1

2 Ny uppgift

During the pre-study phase of a project, the project team is defining its goal. Which of the following objectives is the best example of a SMART goal?
Select one alternative:

- Complete the project within budget and timeline
- Increase sales revenue by 20% in 10 months ✔
- Improve customer satisfaction by collaborating
- Deliver high-quality products

Maximum marks: 1

3 New Question

What is the primary purpose of conducting a pre-study in a project?
Select one or more alternatives:

- [ ] To replace the project idea phase with a concrete plan
- [ ] To decrease uncertainty by mapping assignment, benefits, risks, and stakeholders ✔
- [ ] To start detailed planning and execution immediately
- [ ] To assign all resources before the project proposal is approved

Maximum marks: 1

4 New Question

In the context of pre-study analysis, what does the “100% rule” of the Work Breakdown Structure (WBS) specifically imply?
Select one or more alternatives:

- [ ] Each work package must contribute exactly 100% of the total project cost.
- [ ] Every element of project scope must be captured in the WBS without overlap or omi ✔ on.
- [ ] The project owner must validate 100% of the subgoals before moving forward.
- [ ] All stakeholders must agree to 100% of the listed deliverables before approval.

Maximum marks: 1

5 New Question

During the pre-study, a SWOT analysis is recommended. What is considered a critical limitation if the SWOT is not properly acted upon?

Select one or more alternatives:

- [ ] It automatically biases the project toward defensive planning.
- [ ] It duplicates stakeholder analysis, making it redundant.
- [ ] It becomes a static list without leading to strategies or decisions. ✔
- [ ] It might overemphasize internal weaknesses at the expense of external threats.

Maximum marks: 1

6 New Question

When concluding a pre-study, which condition best ensures alignment with organizational strategy before moving to the planning phase?

Select one or more alternatives:

- [ ] The project goals are defined in SMART terms, but the impact goals remain flexible.
- [ ] The project idea is validated against business goals, with clear delimitations and resource implications. ✔
- [ ] The steering committee has approved a finalized Gantt chart and budget.
- [ ] The WBS has been detailed down to individual tasks to eliminate uncertainty.

Maximum marks: 1

7 New Question

In stakeholder categorisation within a pre-study, what differentiates *core (key) stakeholders* from *primary stakeholders*?

Select one or more alternatives:

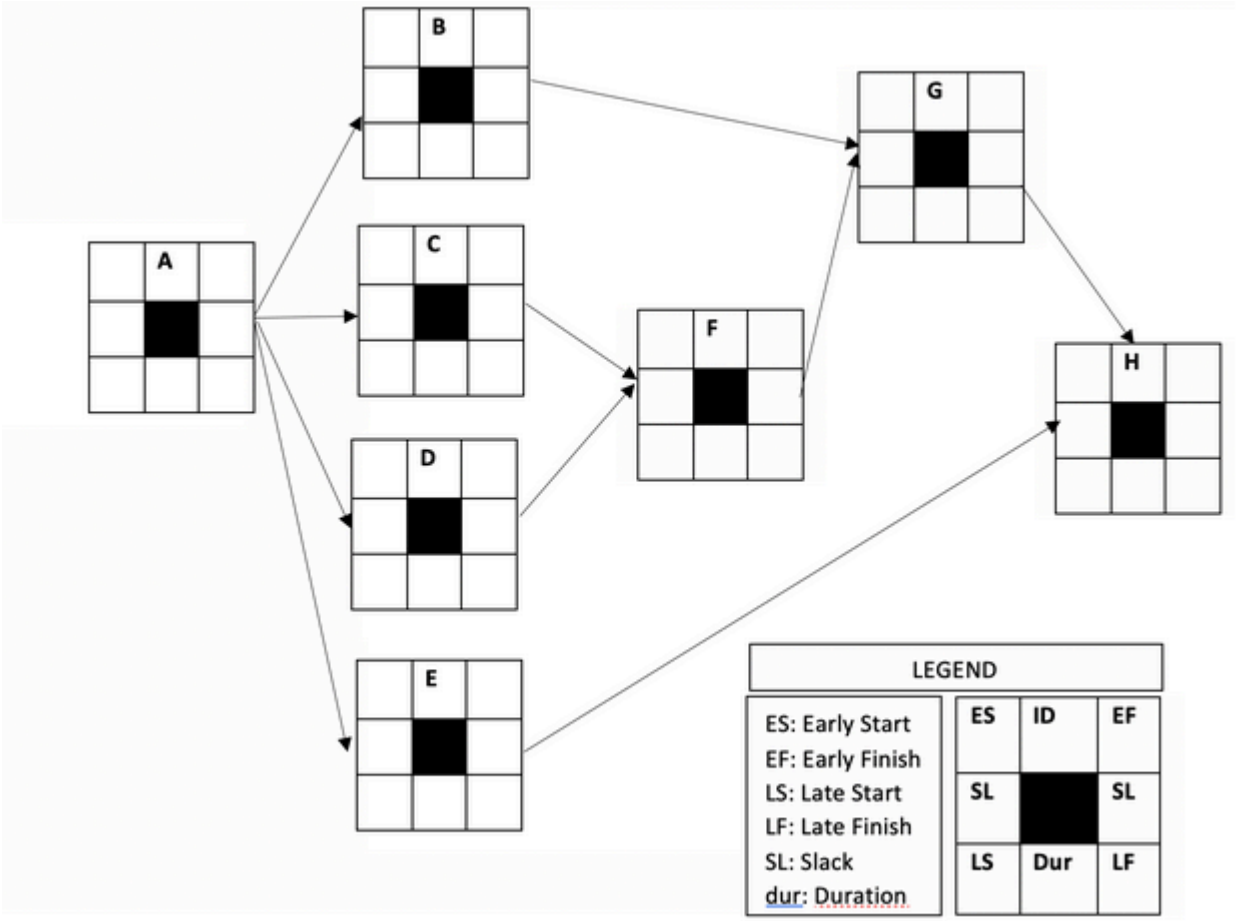
- [ ] Core stakeholders are decision-makers within or directly governing the project, while primary stakeholders are those strongly affected and seeking influence. ✔
- [ ] Core stakeholders are those with financial investments, while primary stakeholders are end-users of deliverables.
- [ ] Core stakeholders require minimal communication, while primary stakeholders require continuous involvement.
- [ ] Core stakeholders are individuals external to the project, while primary stakeholders are internal team members.

Maximum marks: 1

8 New Question

The Project Manager for a new system development project has prepared the activity list with corresponding durations as shown below. The precedence relationships among activities are illustrated in the network diagram provided. Assume the project starts at day 0 **.** Determine The **Critical Path.**

Activity	Duration
A	3
B	8
C	9
D	7
E	25
F	6
G	5
H	4



Select one alternative:

- Critical Path: A → E → H ✔
- Critical Path: A → B → G → H
- Critical Path: A → D → F → G → H
- Critical Path: A →C→F →G→H

Maximum marks: 1

9 New Question

What happens if an activity on the *critical path* is delayed by two days?
Select one alternative:

- The project will be delayed by two days. ✔
- The total slack of the project increases by two days.
- The project completion time is unaffected.
- Non-critical activities will automatically become critical.

Maximum marks: 1

10 New Question

The main purpose of identifying the critical path in a project schedule is to
Select one alternative:

- Identify tasks that must be closely monitored to avoid project delay. ✔
- Estimate total costs for each activity.
- Assign resources to the least time-consuming activities.
- Determine which activities can be safely postponed.

Maximum marks: 1

11 Tools for scope

Which of the following tools would be suitable to present the scope of the project
Select one alternative:

- SWOT analysis
- Work Breakdown Structure ✔
- Iron triangle
- Organizational Breakdown Structure

Maximum marks: 1

12 When is an agile method the right choice

Which of the following are examples of when it is fitting to use agile project methodology?
Select one or more alternatives:

- [ ] When the project owner stops the funding the project after failed pre-study
- [ ] When requirements are very clear to all stakeholders
- [ ] When requirements are very unclear to all stakeholders ✔
- [ ] Once a there is a fixed deadline

Maximum marks: 1

13 New Question

Which of the following statements accurately differentiates a strong matrix from a weak matrix in project management?
Select one alternative:

- ○ In a strong matrix, the functional manager has more authority, while in a weak matrix, the project manager has full control.
- ○ In a strong matrix, the project manager has more authority and control over resourc while in a weak matrix, the functional manager retains most of the control. ✔
- ○ In both strong and weak matrix organizations, the project manager and functional manager share equal authority.
- ○ In a strong matrix, the project manager has less authority, while in a weak matrix, the functional manager has more authority.

Maximum marks: 1

14 Definition Activity plan

Which of the following describes an **activity plan?**

Select one alternative:

- A logical network that shows the order in which the project activities will be handed over
- A logical network that shows the order in which the project activities will be perform ✔
- A logical network that shows the order of SMART goals
- A logical network that shows only the project milestones

Maximum marks: 1

15 What are the 4 stages of team development?

Which of the following description suits well Goal seeking projects
Select one alternative:

- Iterative process and agile project models ✔
- A Well & Clear definied scope
- Enthusiasm, Hope, Panic, Solution.
- Measurable goals set in advance

Maximum marks: 1

16 New Question

A project activity has the following time estimates:

- **Optimistic time (O)** = 4 days
- **Most likely time (M)** = 6 days
- **Pessimistic time (P)** = 10 days

What is the expected time for this activity ?
Select one alternative:

- 6.3 days ✔
- 7.3
- 6 days
- 6.5 days

Maximum marks: 1

17 New Question

What is the primary role of **accrual-based accounting**?
Select one alternative:

- To connects cumulative costs to the customer
- To simplify bookkeeping by delaying cost recognition until the project is completed.
- To provide a financial view that matches costs and revenues with the period when work is actually performed. ✔
- To ensure that project budgets are always balanced at the end of each accounting year, regardless of work progress.

Maximum marks: 1

18 Project status earned value, budget and schedule

You monitored figures on cost and planned/earned value for each individual project week until the data date at the end of the sixth week. The values at the end of the sixth week were:

AC (actual cost) = $200,000
EV (Earned value) = $220,000
PV (planned value) = $210,000.

What is the status of this project at this date?
Select one alternative:

- The project is ahead of schedule and under budget. ✔
- The project is ahead of schedule and over budget
- The project is behind schedule and over budget
- The project is behind schedule and under budget

Maximum marks: 1

19 New Question

During week 8 of a construction project, the following data were collected:

- **Planned Value (PV)** = $400,000
- **Earned Value (EV)** = $360,000
- **Actual Cost (AC)** = $380,000

What is the project's **Schedule Performance Index (SPI) ?**
Select one alternative:

- 105
- 0.95
- 1.11
- 0.90 ✔

Maximum marks: 1

20 New Question

What does the term "Sprint Backlog" mean within the agile method ?
Select one alternative:

- Prioritized tasks to be executed within a shortest time
- Prioritized tasks to be executed within the critical path
- Prioritized tasks to be executed when initial plans are not changes
- Prioritized tasks to be executed within a stage with a defined length ✔

Maximum marks: 1

21 New Question

During the **planning phase**, the project manager begins defining the project organization and assigning roles. What is the **main purpose** of staffing activities at this stage?
Select one alternative:

- To evaluate individual performance for future promotions.
- To recruit all team members permanently into the project organization.
- To delegate daily operational tasks to functional managers.
- To define roles, responsibilities, and competencies required for project execution. ✔

Maximum marks: 1

22 New Question

Project costs are over budget when

Select one alternative:

- CPI = 0
- SPI = 1
- CPI < 1 ✔
- CPI > 1

Maximum marks: 1

23 New Question

A project is a form of work with a specific goal that can be described as a process with a defined beginning and end. Which of the following represents the logical flow according to the stage-gate model?

Select one alternative:

- Initiating, Planning, Executing, Closing ✔
- Initiating, Executing, Planning, Closing
- Initiating, Planning, Closing, Executing
- Planning, Initiating, Executing, Closing

Maximum marks: 1

24 New Question

Which of the following roles suits the "Reference group" in a given project ?
Select one alternative:

- Conduct new assesments
- Form alternative opinions
- Forum for testing ideas ✔
- Lead project changes

Maximum marks: 1

25 New Question

In a project network, Activity A has an Early Start (ES) of 5 days and a Late Start (LS) of 10 days. What is the Total Float for Activity A?
Select one alternative:

- 0 days
- 6 days
- 5 days ✔
- 4 days

Maximum marks: 1

26 New Question

The project owner by contrast to the project manager is (among other things) responsible for:
Select one alternative:

- The project's impact goals ✔
- The project's project goals
- Delegating and following up on activities
- Both the project's project goals and the project's impact goals

Maximum marks: 1

27 New Question

Two projects, **Project A** and **Project B**, each include a critical activity with uncertain duration estimates.

Activity	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Project A	6 days	8 days	10 days
Project B	4 days	8 days	16 days

Which project has **greater uncertainty** in its duration estimate? and hence more riskier
Select one alternative:

- Project A
- Both projects A & B
- Project B ✔
- There is no any risk

Maximum marks: 1

28 New Question

Which leadership style is **most suitable in the early stages of a project** when the team is still forming and roles are unclear?

Select one alternative:

- Laissez-faire style – allowing team members to lead themselves.
- Delegating style – low task, low relationship behavior.
- Participative style – high involvement, shared decision-making.
- Directive style – clear instructions, close supervision, high task focus. ✔

Maximum marks: 1

29 New Question

During closure, what is the main purpose of conducting a project audit?

Select one alternative:

- To update the project schedule for the next phase.
- To identify and punish those responsible for project failures.
- To review only the project's financial results.
- To provide independent assurance of project performance and identify lessons for improvement. ✔

Maximum marks: 1

30 New Question

During project planning, a risk workshop identifies that a key supplier may delay critical components by two weeks. The project manager decides to place an additional order with an alternative supplier in case the delay occurs. Which type of **risk response strategy** does this represent?

Select one alternative:

- Risk transfer
- Risk avoidance
- Risk mitigation ✔
- Risk acceptance

Maximum marks: 1